

数字媒体技术

1. (单选题) Which conversion device can turn sound into corresponding electrical signals?
A.Loudspeaker
B.Sound card
C.Mixing console
D.Microphone
2. (单选题) Which of the following is not one of the three commonly used audio sampling frequencies?
A.11.025 kHz
B.22.05 kHz
C.33.075 kHz
D.44.1 kHz
3. (单选题) From the perspective of digitization, which of the following is not one of the three main factors affecting sound digitization quality?
A.Digital sampling frequency
B.Digital quantization bit depth
C.Number of digital channels
D.Precision of the A/D converter
4. (单选题) Model matching depends on which model library?
A.Speech model library
B.Semantic model library
C.Speaking-rate model library
D.Language model library
5. (单选题) What is the origin of big data?
A.Finance
B.Internet
C.Telecommunications
D.Public administration
6. (单选题) Who is the service object of information visualization?
A.Humans
B.Computers
C.Government agencies
D.Management organizations
7. (单选题) From a macro perspective, which of the following is not a function of data visualization?
A.Information recording
B.Information reasoning and analysis
C.Information cleaning
D.Information dissemination

8. (单选题) A scatterplot matrix uses a set of points in which coordinate system to show relationships between variables?
- A. One-dimensional
 - B. Two-dimensional
 - C. Three-dimensional
 - D. Multi-dimensional

二. 填空题 (共6题, 4.2分)

9. (填空题) The digitization of sound signals consists of sampling, quantization and ____.
10. (填空题) For speech with a bandwidth of 300–3400 Hz, if the sampling frequency is 8 kHz, the quantization bit depth is 8 bits, and mono is used, the uncompressed bit rate is about ____ kbps.
11. (填空题) The number of bit planes of a grayscale image is 1, while a color image has ____ or more bit planes.
12. (填空题) Computer animation is a technology that uses computers to generate a series of continuous pictures for real-time playback. If a film plays 24 frames per second, 2800 frames can be played for about ____ minutes.
13. (填空题) A data compression method that removes details in sound data beyond human hearing ability is called ____ compression.
14. (填空题) The image file format used by digital cameras that supports both lossless and lossy compression is ____.

三. 简答题 (共114题, 90.2分)

15. (简答题) Briefly define digital media. Based on this definition, explain the main differences between digital media and traditional media, and give examples.
16. (简答题) The concept of media convergence was first proposed by Nicholas Negroponte. It is not an independent physical medium. What kind of model does it mainly refer to?
17. (简答题) How does the ITU Telecommunication Standardization Sector classify media?
18. (简答题) Give examples of static media and continuous media.
19. (简答题) Give examples of how emerging technologies such as the Internet, big data, artificial intelligence, cloud computing and mobile communication are applied in media convergence. You may choose one technology or a combination of technologies.
20. (简答题) Briefly describe common network transmission methods used in media convergence technology.
21. (简答题) Briefly describe the main characteristics of digital media and media convergence technology.
22. (简答题) Give examples of applications of media convergence technology in daily life or study, and briefly describe their advantages and disadvantages.
23. (简答题) What is a digital image?
24. (简答题) Briefly describe the purpose of image enhancement and its main methods.
25. (简答题) Briefly describe common color models.

26. (简答题) What types of noise are common in digital image processing?
27. (简答题) What are common methods for digital image denoising?
28. (简答题) Briefly describe the main principle of OCR.
29. (简答题) Briefly describe common application scenarios of multimedia communication.
30. (简答题) Research and summarize the main application fields of remote-sensing image technology.
31. (简答题) Briefly describe the main image and video technologies used in security monitoring.
32. (简答题) What are the main video standards?
33. (简答题) Briefly describe applications of image and video technologies.
34. (简答题) Briefly describe key technologies in digital video processing.
35. (简答题) Briefly define computer vision.
36. (简答题) What are the similarities and differences between digital image processing and computer vision?
37. (简答题) What are the current developments in computer vision?
38. (简答题) Give an example of a typical deep-learning algorithm.
39. (简答题) In what areas is deep learning applied in computer vision? Give an example.
40. (简答题) What are the main steps of the SIFT algorithm?
41. (简答题) Research and summarize the main differences between Fast R-CNN and Faster R-CNN.
42. (简答题) What is the core of stereo vision research?
43. (简答题) Give examples of computer vision applications in daily life.
44. (简答题) What are the main steps of face recognition?
45. (简答题) If the sampling frequency is 44.1 kHz, the quantization bit depth is 16 bits, and the audio is stereo, calculate the amount of digitized sound data per second.
46. (简答题) Briefly define sound intensity and sound pressure, and describe their relationship.
47. (简答题) Briefly explain the concept of the masking effect.
48. (简答题) When two sound sources emit sound at the same time, the measured sound pressure level at a point is 80 dB. After one source is turned off, the measured sound pressure level is 70 dB. What is the sound pressure level of the turned-off source alone?
49. (简答题) Describe the basic characteristics of speech.
50. (简答题) Briefly describe the basic structure of a digital speech communication system.
51. (简答题) Briefly explain the basic concept of speaker recognition.
52. (简答题) Briefly describe the basic principle, applications and main difficulties of speech synthesis.
53. (简答题) Briefly describe the main types of speech coding methods and their advantages and disadvantages.
54. (简答题) Briefly describe the basic process of speech synthesis.
55. (简答题) Briefly define computer graphics and describe its characteristics.

56. (简答题) What are the main geometric elements of graphics? If a house model is to be built, how can its geometric topology be established inside a computer?
57. (简答题) Briefly describe the differences and relationship between computer graphics and computer vision.
58. (简答题) Describe the main characteristics of realistic graphics and non-photorealistic graphics, and briefly compare them.
59. (简答题) Computer graphics is widely used in fields such as CAD, digital art and graphic design. Describe one application around you and explain how it uses graphics technology.
60. (简答题) Visible light is part of the electromagnetic spectrum. Briefly describe its frequency and wavelength ranges.
61. (简答题) Briefly describe the three attributes of color and explain the physical meaning of each.
62. (简答题) In what scenarios is the RGB color model mainly used? Why is it called an additive color system?
63. (简答题) What are the main components of the Phong illumination model?
64. (简答题) Briefly explain the principle of persistence of vision and state the frame rate generally needed for the human eye to perceive continuous animation.
65. (简答题) Briefly describe the principle of keyframe technology.
66. (简答题) Compared with traditional 3D model computer animation, what are the advantages of motion capture technology?
67. (简答题) List the types of virtual reality systems.
68. (简答题) What are the main hardware devices of virtual reality?
69. (简答题) Briefly describe the main features of virtual reality.
70. (简答题) Compared with 3D animation, what are the advantages of virtual reality?
71. (简答题) Briefly explain the concept of virtual reality.
72. (简答题) Briefly describe the main components of a virtual reality system.
73. (简答题) Give examples of applications of virtual reality technology in daily life.
74. (简答题) Briefly describe the main differences between augmented reality and virtual reality.
75. (简答题) Briefly describe the characteristics of digital games.
76. (简答题) According to hardware platforms, what categories can video games be divided into?
77. (简答题) Briefly describe the main impacts of 5G and virtual reality technologies on the game industry.
78. (简答题) The experience of VR games is very different from that of traditional games. In what aspects is this mainly reflected?
79. (简答题) Briefly explain the concept of esports games.
80. (简答题) Give examples of characteristics of esports games compared with other games.
81. (简答题) Briefly describe the main modules of game development and the main function of each module.

82. (简答题) Briefly explain the concept and role of a game engine.
83. (简答题) Briefly describe the main components of a game engine.
84. (简答题) What aspects should be considered when evaluating image compression coding methods?
85. (简答题) Briefly describe the necessity of digital media compression.
86. (简答题) State common coding standards and their typical application scenarios.
87. (简答题) Briefly describe the principle of PCM coding.
88. (简答题) Which audio file formats are lossy compression formats and which are lossless compression formats?
89. (简答题) Briefly describe the characteristics of big data.
90. (简答题) Briefly describe the main steps in the general data-mining process.
91. (简答题) Briefly describe the difference between classification and clustering.
92. (简答题) Briefly describe the main application fields of visual big data.
93. (简答题) Write a common storage-system hierarchy in the order of slow to fast speed, large to small capacity and low to high price.
94. (简答题) Briefly describe the process of host internal I/O access.
95. (简答题) Give examples of the characteristics of data centers.
96. (简答题) Give examples of site-selection requirements for data centers.
97. (简答题) What are the hierarchical layers of a multimedia database? Give examples of application scenarios.
98. (简答题) What categories can cloud computing be divided into by service type? Briefly describe each.
99. (简答题) Briefly define edge computing and introduce its characteristics and application scenarios.
100. (简答题) What are the main applications of modern storage?
101. (简答题) Briefly describe common types of wireless networks.
102. (简答题) Briefly describe the composition principle of a cable television system.
103. (简答题) Briefly describe the development history of mobile communication networks.
104. (简答题) Give examples of applications of 5G ultra-high-definition technology in China.
105. (简答题) Briefly describe the main components of a video conference system and their functions.
106. (简答题) Briefly describe the concept and advantages of triple-play network convergence.
107. (简答题) Research and summarize the main contents of media convergence security.
108. (简答题) Briefly describe the main components of the framework for secure broadcasting in media convergence.
109. (简答题) Briefly describe the types of classical encryption algorithms.
110. (简答题) If the plaintext is "China", what is its corresponding Caesar-cipher ciphertext?
111. (简答题) If the plaintext is "welcome to CUC", use the affine cipher to find the corresponding ciphertext.

112. (简答题) Briefly describe the main characteristics of symmetric encryption algorithms and list typical symmetric encryption algorithms.
113. (简答题) Briefly describe the differences between asymmetric and symmetric cryptographic algorithms.
114. (简答题) Briefly describe the concept and main uses of digital watermarking.
115. (简答题) What categories can audio watermarking algorithms be divided into?
116. (简答题) Briefly describe the concept of biometric authentication and list common biometric authentication technologies.
117. (简答题) Briefly explain the concept of human-computer interaction.
118. (简答题) Briefly describe the development history and current status of human-computer interaction technology.
119. (简答题) What are the main research contents of human-computer interaction?
120. (简答题) What stages has human-computer interaction mainly gone through?
121. (简答题) Briefly describe the concept and characteristics of natural human-computer interaction.
122. (简答题) What are the main types of natural human-computer interaction?
123. (简答题) Briefly explain the concept of multimodal interaction.
124. (简答题) Briefly explain the concept of a human-computer interface.
125. (简答题) Briefly describe the differences among different types of human-computer interaction interfaces.
126. (简答题) Briefly describe the main process of interaction design.
127. (简答题) Research the main interaction functions of Google Glass.
128. (简答题) From the perspective of human-computer interaction, briefly describe the characteristics of Internet television.